

Quantifying Routing Overhead in Quantum Circuit Compilation

A controlled comparison of hardware topology and the hidden effect of physical-qubit labelling

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UNEXPECTED RESULT Physical-qubit labelling can dominate the apparent topology effect. Relabelling the *same* graph moves optimization-level-0 results by up to 15.70× — and one of this study's own conclusions did not survive the control.

How much does restricted qubit connectivity inflate compiled quantum circuits, which interaction patterns are most sensitive to hardware topology — and how much of the measured effect is connectivity rather than how the physical qubits happen to be numbered?

A compiler study, not a physics demonstration: it measures compiler output under modelled connectivity constraints, not QPU fidelity, and makes no quantum-advantage claim.

1,080 canonical compilations — 4 circuit families × 6 sizes (4–24) × 3 topologies at 27 physical qubits × 3 optimization levels × 5 fixed seeds — plus 20,040 in the control: 17,280 relabelling sweep + 1,800 matched baselines + 960 VF2Layout instrumentation. All under Qiskit 2.5.1.

2.140× / 2.304×

Robust topology penalty — line / heavy-hex, levels 1 and 3

15.70×

Largest level-0 shift from relabelling the *same* graph

24 of 24

Relabellings that preserve the QFT / EfficientSU2 ordering

1,080 / 1,080

Canonical results reproduced exactly on a clean rebuild

01 Routing overhead is real

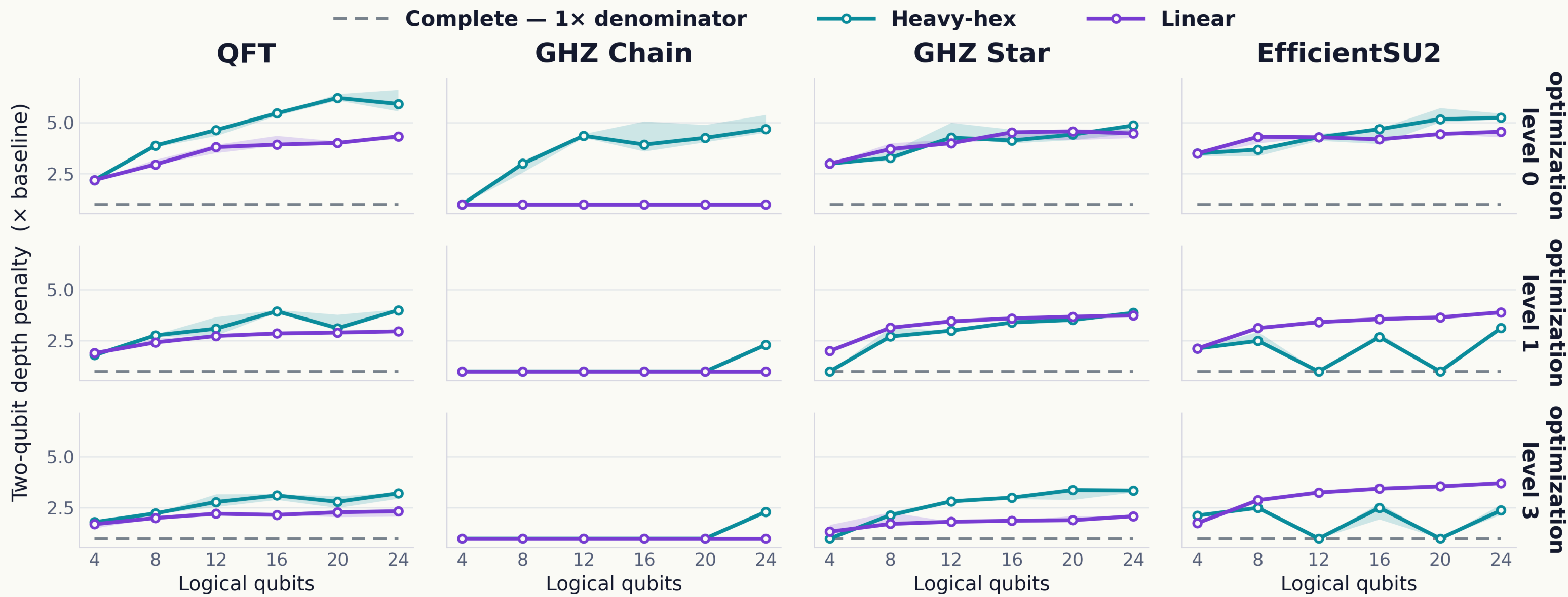
penalty = constrained-topology two-qubit depth ÷ matched complete-connectivity depth · point = median of 5 seeds, band = their IQR

Connectivity costs depth

Excluding level 0, the median penalty is **2.140×** on the line and **2.304×** on heavy-hex. Including it, **2.691×** and **2.909×** — but section 04 shows why level 0 carries no topology claim.

Optimization pays for itself

Level 3 roughly halves the penalty against level 0: **3.904×** → **1.991×** on the line, **4.282×** → **2.174×** on heavy-hex, at about 5× the compile time.

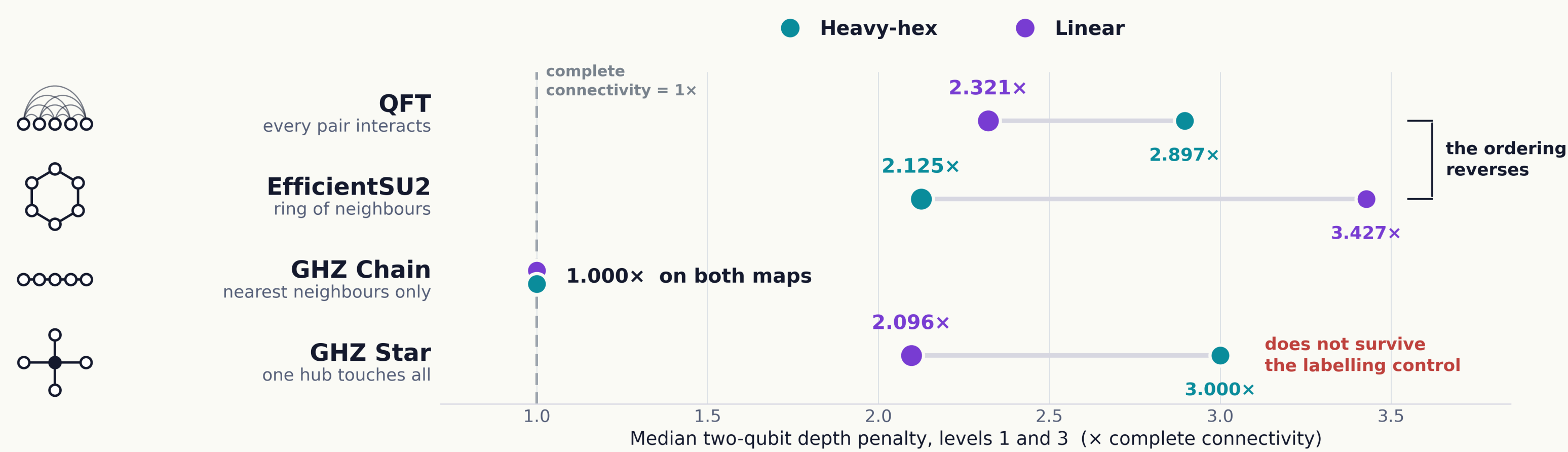


02 Interaction structure decides which topology pays

median over levels 1 and 3, all sizes and seeds · level 0 excluded throughout, for the reason given in section 04

Shape beats sparsity

GHZ chain sits at exactly **1.000×** on both maps — the line already *is* its interaction graph. QFT is cheaper on the line (**2.321×** against **2.897×**) while EfficientSU2 is cheaper on heavy-hex (**2.125×** against **3.427×**). The ordering reverses between two dense circuits on identical hardware.



03 The compiler itself is stochastic

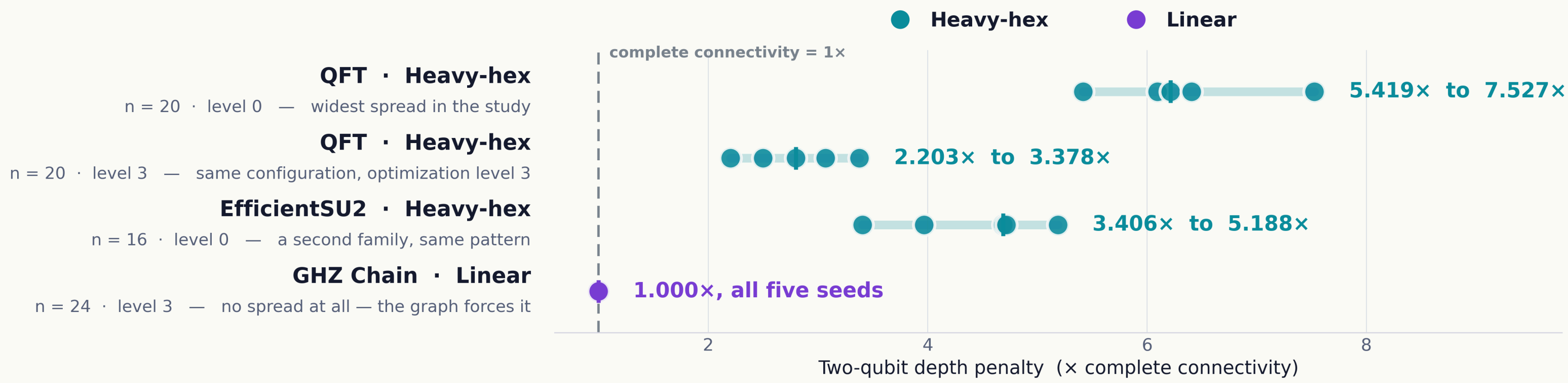
five fixed transpiler seeds (11, 22, 33, 44, 55) · nothing else about the compilation changes between the dots on a row

One circuit, five answers

The widest configuration spans **5.419×** to **7.527×** on seed choice alone. A single-seed benchmark would have reported any point in that range as the result.

It does not vanish with optimization

Level 0 carries the larger *relative* dispersion despite its deterministic layout ($p = 2.6 \times 10^{-6}$), but spread persists at level 3.



04 The control: labelling can dominate the topology effect

Labelling is a confound, and it is measurable

Relabelling the same graph moves level-0 results by up to **15.70×** — about three times the largest topology effect measured anywhere here. This study's labelling is more favourable than every relabelling tested.

But the topology effect survives

At levels 1 and 3 the across-labelling spread collapses to **1.06–1.23×**, and the QFT / EfficientSU2 ordering holds in **24 of 24** relabellings. The mechanism is exact: label-invariant wherever VF2Layout embeds, **28 of 28**.

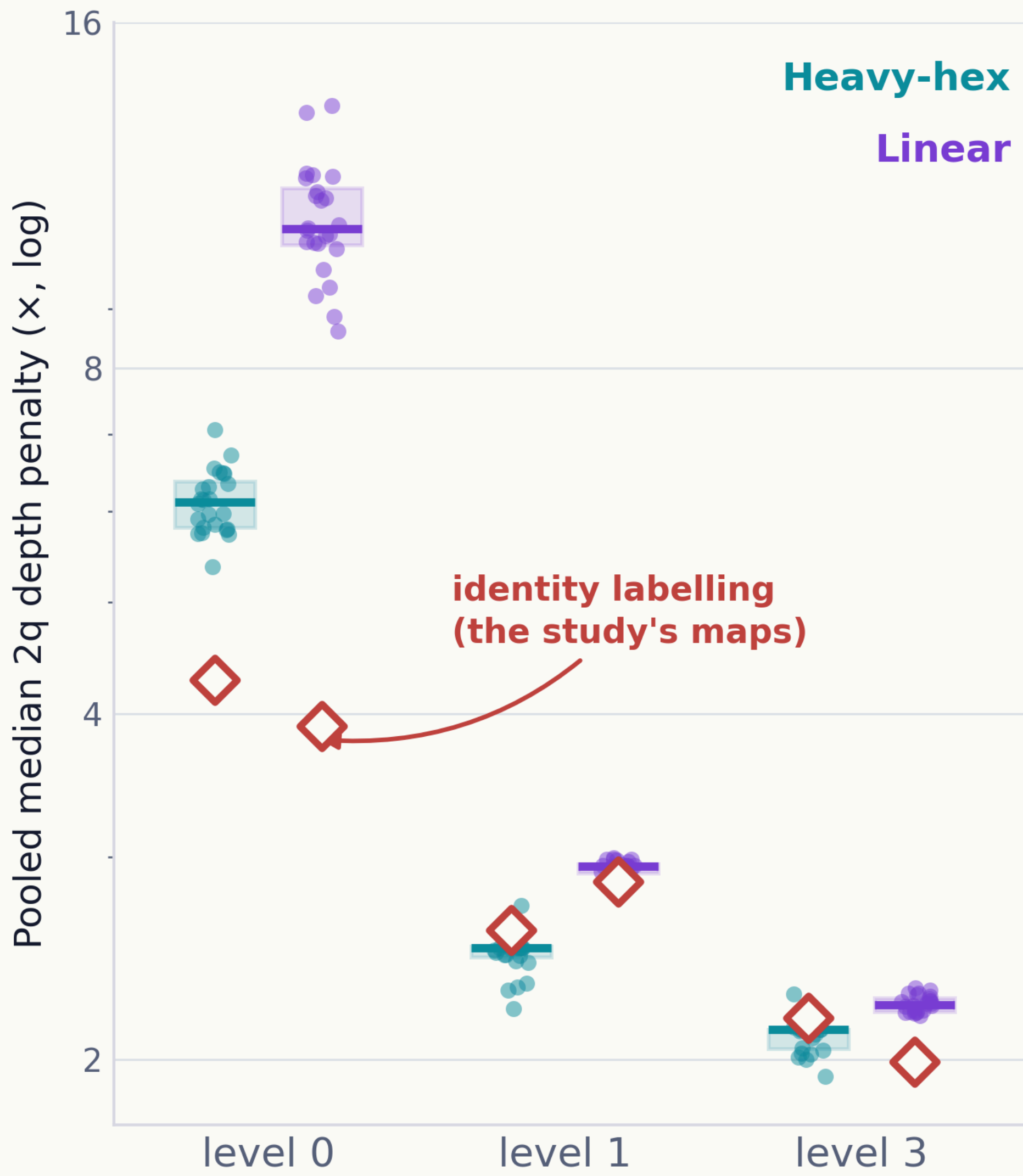
24 deterministically derived relabellings of each constrained map — same graph, same edges, same diameter, only the physical-qubit numbering changes · SHA-256 derivation, pinned by digest, reproducible on any platform

And one claim did not survive

GHZ star looked cheap on the line under this labelling — **1.87×** at level 3, against a relabelling median of **3.99×**. Under relabelling the star is cheaper on heavy-hex in **22 of 24**. Reported as labelling, not connectivity.

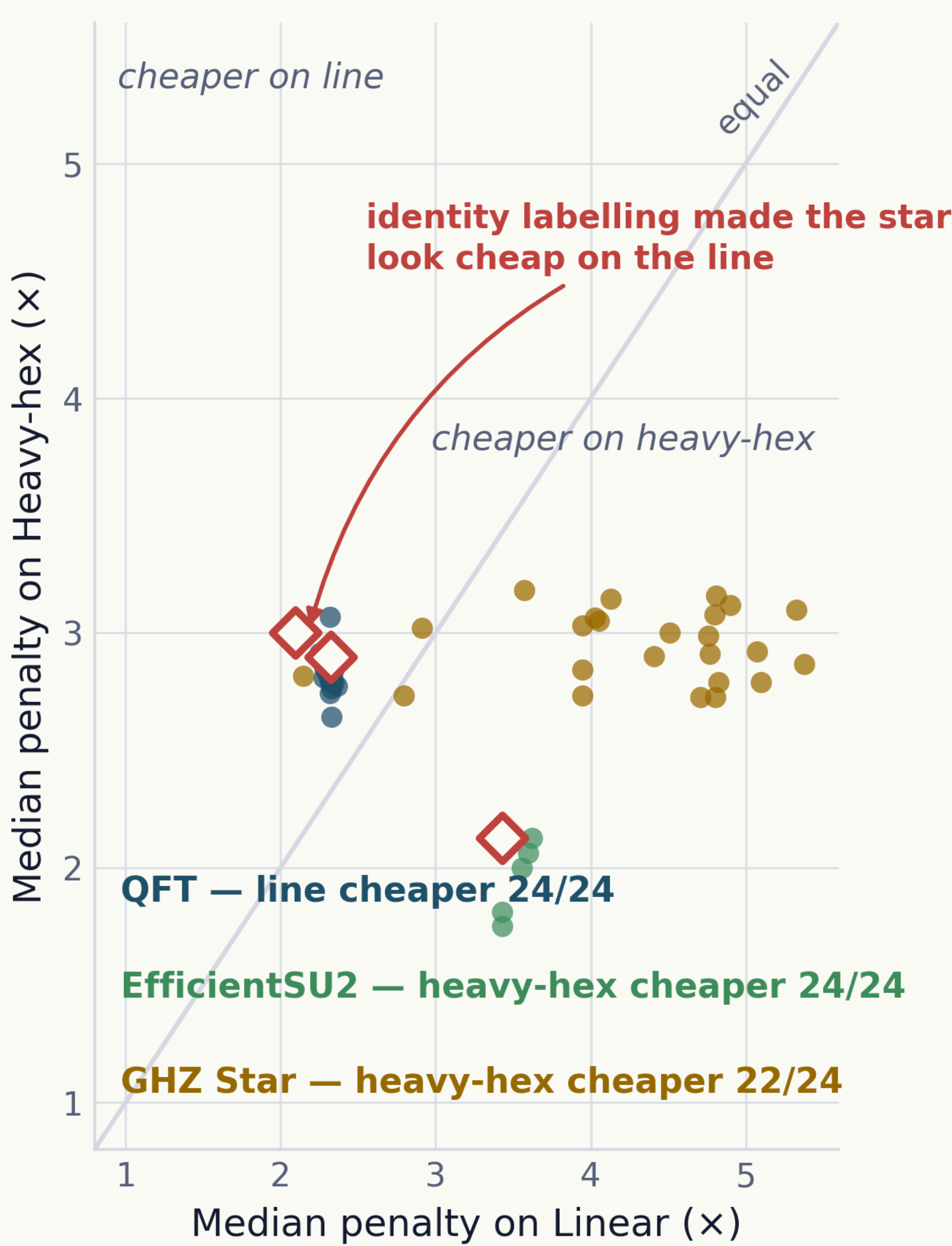
24 relabellings per map

◆ = identity labelling; box = IQR of the 24 relabellings



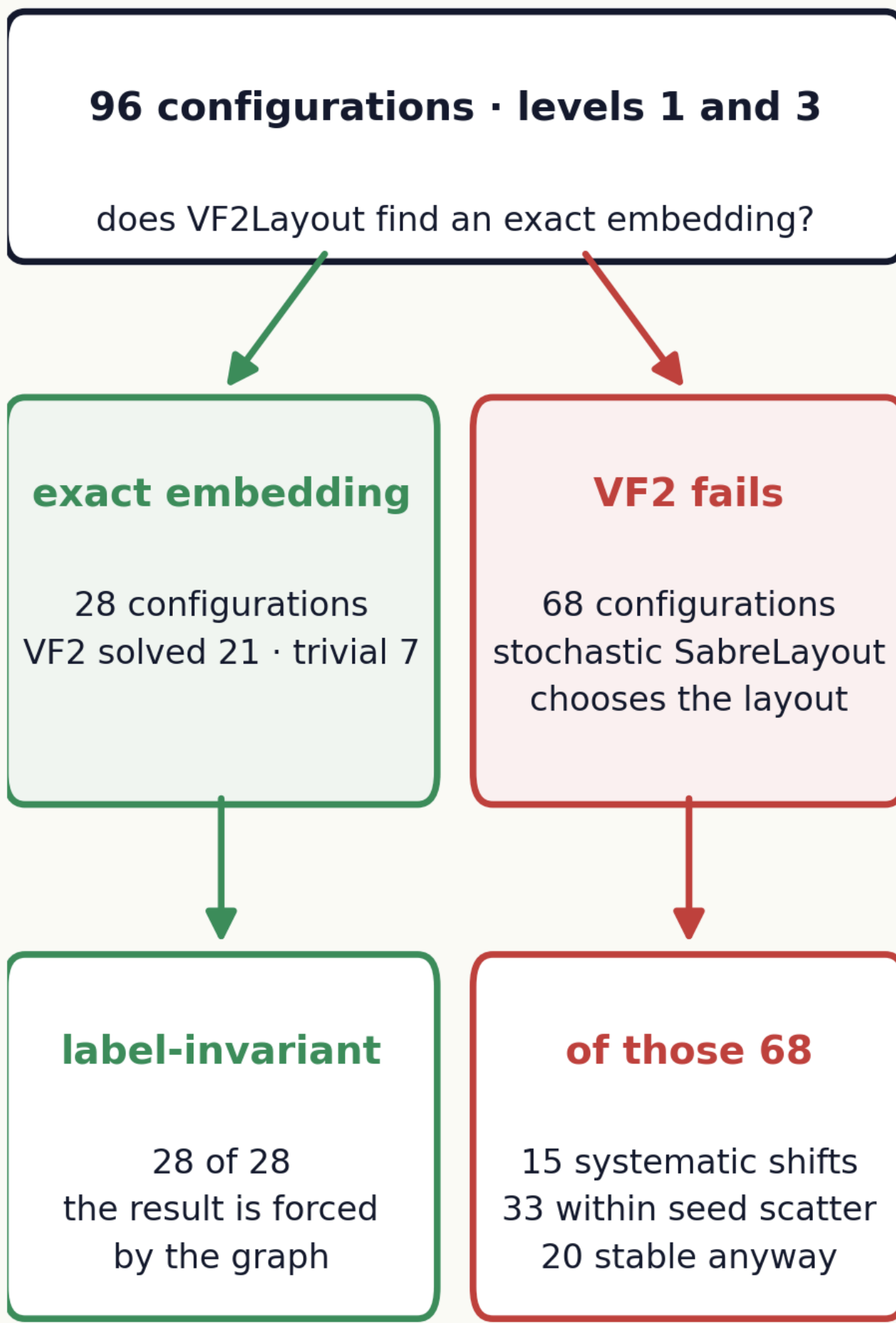
The reversal is label-robust

levels 1 + 3 pooled · one point per relabelling



Why: the layout pass decides

instrumented VF2Layout_stop_reason, 960 compilations



every systematic shift falls in the right-hand branch · 0 exceptions

CONCLUSION

Restricted connectivity does impose measurable routing overhead — **2.140×** on the line and **2.304×** on heavy-hex — but compiler behaviour can confound topology with physical-qubit labelling. Once the level-0 labelling effect is separated out the topology penalty remains, and its circuit-structure ordering holds in **24 of 24** relabellings, while some circuit-specific conclusions become more nuanced.

Limitations

- Compiler-level study, not physical hardware execution.
- One graph per topology class — heavy-hex is Cairo's specific 27-node graph.
- Depends on the tested families, sizes, and the optimization set (0, 1, 3).
- Five seeds and 24 relabellings characterise, but do not exhaust, either space.

Where level 0 stands

Level 0 is a declared Qiskit preset and no stored value is changed, but it pins the layout to the identity permutation, so it measures numbering as much as connectivity. Every topology claim on this poster excludes it. Where an exact embedding exists the result is label-invariant, **28 of 28**; every systematic shift falls where VF2Layout fails, **15 of 15**. No exceptions in either direction.

Reproducible benchmark

Python 3.12 · Qiskit 2.5.1 · fixed seeds · locked dependencies. Recompiling all 1,080 canonical points reproduced the stored metrics **exactly, 1,080 of 1,080**. Sweep conditions derive from a SHA-256 keystream, pinned by digest in the test suite.

github.com/ayujos/quantum-routing-overhead

